

✓ PDS PRACTICAL 2

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Aim : Data Ingestion and Cleaning: Acquire data and prepare it for analysis.

Theory

✓ 1. What is data ingestion and why is it important? Which Python library is used?

Answer:

Data ingestion is the process of collecting, importing, and loading data from various sources into a system where it can be stored, processed, and analyzed. These sources can include structured data (CSV files, databases), semi-structured data (JSON, XML), and unstructured data (logs, images).

In data science, data ingestion is the first and foundational step of the data pipeline. If this step is poorly handled, every subsequent step—cleaning, analysis, and modeling—will be flawed.

Importance of Data Ingestion:

- Ensures data is available in a usable and structured format.
- Integrates data from multiple sources into one system.
- Maintains consistency and completeness of data.
- Enables real-time or batch data processing.
- Forms the base for accurate analysis and machine learning.

In Python, the most commonly used library for data ingestion and cleaning is Pandas. It provides powerful functions like:

- `read_csv()` for CSV files
- `read_excel()` for Excel files
- `read_json()` for JSON data

Example:

```
import pandas as pd
df = pd.read_csv("titanic.csv")
```

2. Why is data cleaning necessary?

Answer:

Data cleaning is the process of identifying and correcting errors, inconsistencies, and missing values in a dataset before performing analysis or applying machine learning algorithms.

Real-world data is rarely clean. It often contains:

- Missing values (e.g., unknown age in Titanic dataset)
- Duplicate records
- Incorrect or inconsistent formats
- Outliers or noise

If data is not cleaned:

- Analysis results become inaccurate or misleading.
- Machine learning models learn wrong patterns.
- Predictions become unreliable.

Benefits of Data Cleaning:

- Improves data quality and consistency.
- Increases accuracy of analysis.
- Enhances performance of machine learning models.
- Reduces bias caused by incorrect data.

Example (Titanic dataset): Missing values in the `Age` column must be handled:

3. How do you load a CSV file into a Pandas DataFrame?

Answer:

A CSV (Comma-Separated Values) file is one of the most common formats for storing tabular data. In Python, it can be loaded into a Pandas DataFrame using the

`read_csv()` function.

This function reads the file and converts it into a structured format where:

- Rows represent observations.
- Columns represent features.

Basic Syntax:

```
import pandas as pd
df = pd.read_csv("titanic.csv")
```

Optional Parameters:

- `head()` → to preview data
- `index_col` → to set index column
- `na_values` → to define missing values

Example:

```
df = pd.read_csv("titanic.csv")  
df.head()
```

4. What is the difference between `head()` and `tail()`?

Answer:

Both `head()` and `tail()` functions are used to quickly inspect the dataset, but they differ in which part of the data they display.

Function	Description
<code>head()</code>	Displays the first few rows (default = 5)
<code>tail()</code>	Displays the last few rows (default = 5)

Purpose:

- `head()` helps understand the structure and column names.
- `tail()` helps verify the end of the dataset and detect anomalies.

Example:

```
df.head()  
df.tail()
```

You can also specify the number of rows:

```
df.head(10)  
df.tail(3)
```

5. How can you identify missing values in a dataset?

Answer:

Missing values are common in real-world datasets and must be identified before cleaning. In Pandas, missing values are typically represented as `NaN` (Not a Number).

You can detect missing values using the following methods:

1. `isnull()` or `isna()`

Returns a Boolean DataFrame indicating missing values.

```
df.isnull()
```

2. Count missing values using `sum()`

```
df.isnull().sum()
```

This shows how many missing values are present in each column.

3. Check if any missing values exist

```
df.isnull().any()
```

4. Visual inspection

```
df.info()
```

This shows non-null counts for each column.

Example Insight (Titanic dataset):

- `Age` and `Cabin` columns often contain missing values.
- These must be handled before analysis.

1. What is data ingestion and why is it important in data science? Which Python library is commonly used for data ingestion and cleaning?

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

```
url = '/content/train (1).csv'
df = pd.read_csv(url)
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 211
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs)	female	38.0	1	0	PC 175

```
df.isnull().sum()
```

	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2

dtype: int64

```
age_column = df['Age']
```

```
age_column.head()
```

	Age
0	22.0
1	38.0
2	26.0
3	35.0
4	35.0

dtype: float64

```
df['date'] = '14 Apr 1912'
```

```
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 211
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briass	female	38.0	1	0	PC 175

```
sorted_df = df.sort_values(by='Fare', ascending=False)
```

```
sorted_df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket
679	680	1	1	Cardeza, Mr. Thomas Drake Martinez	male	36.0	0	1	PC 1775
258	259	1	1	Ward, Miss. Anna	female	35.0	0	0	PC 1775

```
sorted_df = df.sort_values(by='Fare', ascending=False)
```

```
sorted_df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket
679	680	1	1	Cardeza, Mr. Thomas Drake Martinez	male	36.0	0	1	PC 1775
258	259	1	1	Ward, Miss. Anna	female	35.0	0	0	PC 1775

```
df_renamed = df.rename(columns={
    'Sex': 'Gender',
    'Pclass': 'Passenger_Class'
})
```

```
df_renamed.head()
```

	PassengerId	Survived	Passenger_Class	Name	Gender	Age	SibSp	Parch
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs)	female	38.0	1	

```
df.describe()
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000

```
filtered_df = df[(df['Age'] > 30) & (df['Fare'] > 50)]
```

```
filtered_df.head()
```

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticke	
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	P 1759
				Futrelle, Mrs.					

```
filtered_df = df[(df['Age'] > 30) & (df['Fare'] > 50)]
```

```
filtered_df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticke
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	P 1759
				Futrelle, Mrs.					

```
grouped = df.groupby('Sex')['Fare'].mean()
```

```
grouped
```

	Fare
Sex	
female	44.479818
male	25.523893

```
dtype: float64
```



```
df2 = df[['PassengerId', 'Survived']]

merged_df = pd.merge(df, df2, on='PassengerId')

merged_df.head()
```

	PassengerId	Survived_x	Pclass	Name	Sex	Age	SibSp	Parch	Ti
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 2
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briags	female	38.0	1	0	PC 1

```
df_dropped = df.drop(columns=['Cabin'])

df_dropped_rows = df.dropna(subset=['Age'])

df_dropped.head()
df_dropped_rows.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Tick
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 211
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briags	female	38.0	1	0	PC 175

```
df.to_csv("processed_titanic.csv", index=False)

print("File saved successfully!")

File saved successfully!
```

Conclusion

This assignment demonstrated how to perform data ingestion, cleaning, and basic analysis using Pandas. The dataset was loaded and processed using operations like filtering, sorting, grouping, and handling missing values. Statistical measures such as mean and median helped in understanding the data distribution.

Through these steps, it became clear that raw data requires proper preprocessing before analysis. Basic data analysis also helped in identifying patterns and relationships within the dataset. Overall, this assignment provided a strong foundation for further data analysis and machine learning tasks.